

Benjamin Seguin

Paris, France

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EDUCATION

Université PSL — ENS, Dauphine, Mines

Paris, France

M.Sc. in Computer Science (IASD track)

Sep. 2025 – Sep. 2026

- Relevant coursework: Mathematics of Deep Learning (B. Loureiro), Dimension Reduction and Manifold Learning (E. Aamari), Non-convex Inverse Problems (I. Waldspurger), Foundations of Machine Learning (F. Bach), Optimization for Machine Learning (C. Royer).
- Cumulative Average: 15/20, Master's thesis: 19/20.

HEC Montréal

Montréal, Canada

M.Sc. in Financial Engineering, Thesis track

Sep. 2023 – Dec. 2024

- GPA 4.0/4.3; *mention d'excellence*; Excellence Scholarship.
- Master's thesis (4.3/4.3, *mention d'excellence*): *Strategies to Monetize Sentiment Extracted from Earnings Call Transcripts Using NLP and LLMs*.
- Teaching Assistant in Time Series Econometrics.

HEC Montréal

Montréal, Canada

B.B.A. in Finance, Bilingual track

Sep. 2018 – Jun. 2022

RESEARCH EXPERIENCE

LAMSADE, MILES team — Université Paris Dauphine-PSL

Paris, France

Research Intern, supervised by Prof. Clément Royer

Apr. 2026 – Sep. 2026

- Complexity analysis of a hybrid algorithm combining Nonlinear Conjugate Gradient with Negative Curvature Descent for non-convex optimization. Derived per-regime iteration complexity bounds on strict saddle functions, including novel rates in the local strong convexity regime.

HEC Montréal — Computational Finance

Montréal, Canada

Research Assistant, supervised by Prof. David Ardia

Jan. 2025 – Sep. 2025

- Developed three open-source R packages for econometric inference (regime-switching correlations, mean-variance spanning tests, luck-corrected peer performance). Three co-authored preprints on SSRN (Ardia & Seguin, 2025).

PROFESSIONAL EXPERIENCE

Caisse de Dépôt et Placement du Québec (CDPQ)

Montréal, Canada

Quantitative Research Intern

Jun. 2024 – Dec. 2024

- Built ML pipelines for LLM-based sentiment extraction (reduced inference time 25min → 2min); developed OOP architecture for low-latency quantitative signal computation.

TECHNICAL SKILLS

- **Programming:** Python (PyTorch, JAX, NumPy, Pandas, scikit-learn), R, $\text{\LaTeX}$, Git.
- **Mathematics:** Linear algebra, optimization, probability, stochastic calculus, time series.
- **Languages:** French (native), English (fluent).